1 THE DATA REALITY

OpenAlex civil-engineering corpus

title = summary field · abstract = background field

record with abstract

- title
- abstract

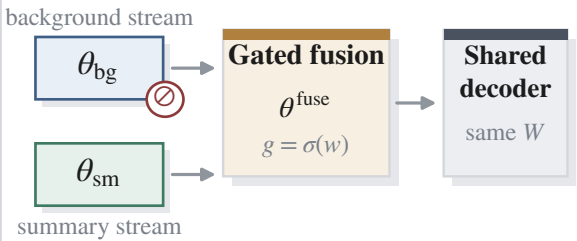
title-only record

- title

abstract missing

The missing side changes the observation, not the modelling goal

2 FIELD-COOPERATIVE MODELING



field masking: drop a side, keep the model

Fields co-train through one shared topic–word matrix and one vocabulary

no separate title-model / abstract-model

3 ONE MODEL, THREE REGIMES

complete record

θ_{sm}, θ_{bg} → same topics

abstract missing

θ_{sm} + fallback → same topics

title-only record

θ_{sm} only → same topics

unified topic–word read-out (β / class TF–IDF)

screening 7,102 civil articles

with topics, not keywords